

Joel Tchouke

📍 Mankato MN ✉ tchoukejoel@gmail.com ☎ 929-339-7034 📁 Portfolio in LinkedIn 🐙 GitHub

Technical Skills

Languages: C, C++, RUST, Python, Bash, VHDL, SystemVerilog, TypeScript, SQL

Systems & Embedded: Embedded Linux, microcontrollers, firmware development, Hardware/Software Integration, Sensor Interfacing, System Debugging, FPGA, ROS, RTOS, MQTT

Tools & Platforms: Git/GitHub, Linux, MATLAB, KiCad, Logic Analyzers, Structured Logging, Automated Testing

Engineering Areas: Control Systems, Signal Processing, Embedded Systems Design, System Integration, Networking Fundamentals

Experience

Embedded Software Engineer Intern

Fargo, ND May 2025 – Dec 2025

AGCO

- Implemented firmware modules in C/C++ for ARM-based control systems, improving subsystem reliability through structured debugging and instrumentation
- Built repeatable validation workflows to test hardware interfaces and embedded software behavior across multiple subsystems
- Converted MATLAB algorithm models into embedded implementations, reducing validation time by **40%**
- Diagnosed timing and synchronization defects across interacting firmware modules through controlled experiments and signal tracing

Cybersecurity Analyst

Mankato, MN Jan 2025 –

Minnesota State University IT Solutions

Present

- Automated investigation workflows using Python and structured log analysis, reducing manual analysis time by **30–50%**
- Correlated telemetry across network and host systems to identify abnormal device behavior and root causes of incidents
- Produced structured technical reports and reproducible investigation steps to support engineering response teams

Projects

Innovision — Embedded Smart Glasses Platform

- Designed an embedded Linux system integrating computer vision, audio processing, and wireless control modules
- Implemented text-to-speech functionality and device communication interfaces enabling assistive interaction workflows
- Integrated camera input, object recognition, and audio feedback pipelines to deliver responsive real-time interactions
- Profiled concurrency bottlenecks and optimized multimedia pipelines, reducing response latency by **30%**

ORBS — Autonomous Embedded Drone System

- Developed an autonomous drone platform integrating embedded control systems, acoustic sensors, and onboard processing
- Implemented signal processing and adaptive noise mitigation algorithms using real-time sensor inputs
- Designed feedback control loops to stabilize system response and maintain consistent operation during dynamic flight conditions
- Built hardware and software interfaces for sensor acquisition, telemetry collection, and system diagnostics

Education

Minnesota State University, Mankato

Jan 2022 – Dec 2026

B.S. Computer Engineering — Physics Minor — Honors Program

GPA: 3.7/4.0

- Coursework: Control Systems, Embedded Systems, Computer Architecture, Digital Logic, Electronics, Signals and Systems
- Codepath A110: Applications of AI Engineering
- Dean's List (Fall 2022 – Fall 2024)

Leadership

IEEE Student Branch — Treasurer

Managed organizational budgets and financial reporting for engineering events and technical initiatives

MavPASS Leader — Electrical Engineering Tutor

Led collaborative learning sessions and technical instruction for engineering coursework

Cybersecurity Association — President

Organized technical workshops, security competitions, and industry speaker events to grow student engagement in cybersecurity; led the team to **3rd place** in the Midwest Collegiate Cyber Defense Competition (CCDC)